

SPECTRAVIDEO USERS GROUP

WELLINGTON

NEWSLETTER

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Meetings are held on the third Monday of the month at 7.30 pm in the Staff Room of Newlands Intermediate, Bracken Road, Newlands, Wellington. Attendance currently averages 10-15 with ideas, information, problems and matters of interest being shared.

NEXT MEETING MAY 16th

COMMITTEE MEMBERS

Chairman	Jim Oliver	ph 699 618
Secretary/Treasurer	Ross Fletcher	ph 788 454
Librarian	Scotty Baxter	ph 784 619

Public Domain BASIC Jim Oliver

Public Domain CP/M Ross Fletcher

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### C O N T E N T S

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From your Editor,

Hi to you all from me in this my first issue!! What fun heigh ho! Having not done this sort of thing before you will have to take what you get for a while and then you will just have to take what you get!

My thanks to Doug Pratt, new member but old hand with SVI, ex Dunedin. Doug is handy with a soldering iron and came to my rescue so this could be produced on the genuine article rather than one of those hideous unmentionables that I am so reluctantly forced to use at work. Having a double sided 80 track per side Shugart in B has never done my power supply much good. The original diodes cooked some annums ago and one Steve Lacey, now of parts unknown but who owes me lots of newsletter material, bunged a couple of new trans in and a whacking great bridge resistor. Best screen display I've ever seen on an SVI 80 col.

Lasted 8 months before I had to relace it because of overheating. The new one went the same way in approximately the same time - along comes Doug, technician extraordinaire. The symptom was unscheduled automatic reboots - when the power to the expander didn't simply disappear. In went a new bridge, but this time mounted on the back wall of the chassis as a heat sink with pretty pieces of coloured wire filched from the mains cord off my refrigerator reaching into the PCB- deadlines must be met and the middle name of any editor must be "Sacrifice"! Fortunately the fridge is defunct anyway. The moral of the story however is not that you should avoid modifications - I could not do without my 664 K bytes free on Drive B - but to look past the obvious when trouble occurs. Putting things back together Doug's sharp eye spotted a loose soldered joint on the 12volt power lead socket. While heat from the bridge was obviously getting out of hand it was probably the faulty solder join (somewhat scorched also) which actually let go of the power supply causing the auto reboots.

Can I suggest an inspection from time to time of all vulnerable areas where physical stress or heat may cause weakening of solder or and sockets etc. Some of our machines are now advancing in age and have given many hours of use. A little preventative maintenance might not go astray. However I am sure you will, like me, wait until it actually stops working - on your own head be it!

This newsletter is produced using Flashprint which is a commercial program to allow full access to your printer from Wordstar. If your printer can do it, so can Flashprint. Further information is available from me or Jim Oliver.

In the next issue I would like to publish a list of current members and addresses. If anyone has a problem with this please let me know.

Finally, keep those articles rolling in - there are still lots of good things happening in the world of SPECTRAVIDEO!

.....John

```

10 |*****
20 |*
30 |*          CMDPEEK/POKE/LINE          *
40 |*          BY                          *
50 |*          BOB CURWOOD                 *
60 |*          CHRISTCHURCH               *
70 |*
80 |*****
90  ORG 0D503H
100 LOAD ONH
110 GETBYT: EQU 1AA6H
120 OFFSET: EQU 0F000H
130 SNERR: EQU 0BEDH
140 POKTOK: EQU 152
150 PEKTOK: EQU 151
160 LNETOK: EQU 0AFH
170 DSHTOK: EQU 0F4H
180 ALTHOK: LD A,0C3H
190          LD (OFF9CH),A
200          LD HL,START
210          LD (OFF9DH),HL
220          RET
230 SAVEZ:  DB '*'
240 START:  LD A,(HL)
250          CP POKTOK
260          JR Z,POKE
270          CP LNETOK
280          JR Z,BOX
290          CP 255
300          RET NZ
310          RST 10H
320          CP PEKTOK
330          RET NZ
340 PEEK:   INC HL
350          CALL LOCATE
360          PUSH HL
370          CALL MULTI
380          POP HL
390          PUSH HL
400          LD HL,(SAVLOC)
410          CALL CHGBNK
420          LD E,(HL)
430          CALL CHGBNK
440          LD A,32
450          ADD A,E
460          LD (SAVEZ),A
470          EI
480 EREXIT: POP HL
490 EXIT:  POP AF
500          RET
510 POKE:  INC HL
520          CALL LOCATE
530          PUSH HL
540          CALL MULTI
550          POP HL
560          LD A,(HL)
570          PUSH HL
580          RST 0H
590          DB ','
600          POP HL

```

IUSED BY PEEK & BOX  
 ICMD MUST BE FOLLOWED BY EITHER  
 IPOKE, PEEK OR LINE  
 IIF NONE OF THOSE THREE  
 ITHEN RETURN TO ILLEGAL FUNCTION  
 IGET THE BYTE  
 ICONVERT TO NORMAL ASCII  
 ISTORE IT HERE  
 IERROR EXIT FROM SUBROUTINES NEED AN  
 IEXTRA POP  
 ILOCATE UPDATES SAVEX & SAVEY  
 IMULTI CONVERTS THEM TO THE ACTUAL  
 IRAM LOCATION PLUS F000  
 ITHIS IS THE 2ND COMMA

```

610      INC HL
620      CALL GETBYT      ;THIS IS THE LAST PARAMETER
630      CP ' '
640      RET C
650      SUB 32
660      PUSH HL
670      LD E,A
680      CALL CHGBNK
690      LD A,E
700      LD HL, (SAVLOC)
710      LD (HL),A
720      CALL CHGBNK
730      POP HL
740      EI
750      JR EXIT
760 BOX:  INC HL
770      LD A, (HL)
780      CP '('
790      RET NZ
800      INC HL
810      CALL LOCATE      ;DONT NEED SAVEX/SAVEY THIS TIME
820      PUSH HL
830      CALL MULTI      ;DO NEED SAVLOC THO'
840      POP HL
850      LD A, (HL)
860      CP ')'
870      RET NZ
880      INC HL
890      LD A, (HL)
900      CP DSHTOK
910      RET NZ
920      INC HL
930      LD A, (HL)
940      CP '('
950      RET NZ
960      INC HL
970      CALL LOCATE      ;THIS TIME IT'S SAVEX/SAVEY that are
980      LD A, (HL)      ;NEEDED
990      CP ')'
1000     RET NZ
1010     RST 10H
1020     INC HL
1030     JR Z,PVIOUS
1040     CP ','
1050     RET NZ
1060     CALL GETBYT
1070     CP ' '
1080     RET C
1090     LD (SAVEZ),A
1100 PREVIOUS:  PUSH HL      ;NEXT COMMAND
1110     CALL CHGBNK
1120     LD A, (SAVEZ)
1130     SUB 32
1140     LD HL, (SAVLOC)
1150     LD DE,80
1160     LD BC, (SAVEY)      ;CHARS: PER LINE
1170     INC B                ;X IN B,Y IN C
1180     INC C
1190     PUSH HL
1200     PUSH BC

```

```

1210 TOPLIN: LD (HL),A
1220 INC HL
1230 DJNZ TOPLIN
1240 DEC HL
1250 ROTLIN: LD (HL),A
1260 ADD HL,DE
1270 DEC C
1280 JR NZ,ROTLIN
1290 POP BC
1300 POP HL
1310 LFTLIN: LD (HL),A
1320 ADD HL,DE
1330 DEC C
1340 JR NZ,LFTLIN
1350 OR A
1360 SBC HL,DE
1370 BOTLIN: LD (HL),A
1380 INC HL
1390 DJNZ BOTLIN
1400 CALL CHGBNK
1410 EI
1420 EREXIT1: POP HL
1430 EXIT1: POP AF
1440 RET
1450 SAVEY: DB 00
1460 SAVEX: DB 00
1470 SAVLOC: DW 0000
1480 TOGGLE: DB 0
1490 LOCATE: CALL GETBYT
1500 CP 80
1510 JR NC,EREXIT1
1520 LD (SAVEY),A
1530 LD A,(HL)
1540 CP ','
1550 JR NZ,EREXIT1
1560 INC HL
1570 CALL GETBYT
1580 CP 24
1590 JR NC,EREXIT1
1600 LD (SAVEY),A
1610 RET
1620 MULTI: LD BC,850H
1630 LD A,(SAVEY)
1640 LD E,A
1650 LD D,0
1660 LD HL,0
1670 SAVLP: SRL C
1680 JR NC,BYPASS
1690 ADD HL,DE
1700 BYPASS: SLA E
1710 RL D
1720 DEC B
1730 JR NZ,SAVLP
1740 LD DE,OFFSET
1750 ADD HL,DE
1760 LD A,(SAVEY)
1770 LD E,A
1780 LD D,0
1790 ADD HL,DE
1800 LD (SAVLOC),HL

```

ITO FAR TO 1ST EXIT FOR RELATIVE JUMP

ITHESE TWO MUST BE IN THIS ORDER SO  
IBC CAN PICK THEM UP CORRECTLY

IMUST BE ZERO TO START WITH  
IThis SUBROUTINE TAKES 2 PARAMETERS  
IFROM THE BASIC PROGRAM, SEPARATED BY  
IA COMMA, DECODES THEM AND STORES X IN  
ISAVEY & Y IN SAVEY. X IS IN THE RANGE  
10 TO 79, Y IS IN THE RANGE OF 0 TO 23  
INEGATIVE INTEGERS AND THOSE OUTSIDE  
ITHESE RANGES WILL GIVE ILLEGAL FUNCTION  
ICALL.

```

1810          RET
1820 CHGBNK:  DI          ;TOGGLE IS TOGGLED (!) BETWEEN 0
1830          LD A,(TOGGLE) ; & 255! IS NORMAL RAM, 255 IS
1840          XOR 255       ;80 COL: RAM. TOGGLE MUST BE CALLED
1850          LD (TOGGLE),A ;TWICE, INTERRUPTS MUST BE DISSABLED
1860          OUT (88),A    ;WHILE 80 COL: RAM IS ENABLED.
1870          RET
1880 WRKEND:  EQU *
1890 END
    
```

PAGE 1

|         |             |             |             |      |
|---------|-------------|-------------|-------------|------|
| ALTHOK  | D503 BOX    | D566 BOTLIN | D5C5 BYPASS | D601 |
| CHGBNK  | D617 DSHTOK | 00F4 EREXIT | D538 EXIT   | D53C |
| EREXIT1 | D5CD EXIT1  | D5CE GETBYT | 1AA6 LNETOK | 00AF |
| LFTLIN  | D5BD LOCATE | D5D5 MULTI  | D5F0 OFF8ET | F000 |
| POKTOK  | 0098 PEKTOK | 0097 PEEK   | D520 POKE   | D53E |
| PREVIUS | D59A ROTLIN | D5B6 SNERR  | 08ED SAVEZ  | D50F |
| START   | D510 SAVEY  | D5D0 SAVEX  | D5D1 SAVLOC | D5D2 |
| SAVLP   | D5FC TOPLIN | D5B1 TOGGLE | D5D4 WRKEND | D623 |

```

D503 3E C3 32 9C FF 21 10 D5 22 9D FF C9 2A 7E FE 98 >.2...!..."#~..
D513 28 29 FE AF 28 4D FE FF C0 D7 FE 97 C0 23 CD D5 ()..(M.....N..
D523 D5 E5 CD F0 D5 E1 E5 2A D2 D5 CD 17 D6 5E CD 17 .....#.....^..
D533 D6 3E 20 83 32 0F D5 FB E1 F1 C9 23 CD D5 D5 E5 > .2.....N.....
D543 CD F0 D5 E1 7E E5 CF 2C E1 23 CD A6 1A FE 20 D8 .....~...W.....
D553 D6 20 E5 5F CD 17 D6 7B 2A D2 D5 77 CD 17 D6 E1 . _...{#.W.....
D563 FB 18 D6 23 7E FE 28 C0 23 CD D5 D5 E5 CD F0 D5 ...W~.(.N.....
D573 E1 7E FE 29 C0 23 7E FE F4 C0 23 7E FE 28 C0 23 ~.~.N~..N~.(.N
D583 CD D5 D5 7E FE 29 C0 D7 23 28 0C FE 2C C0 CD A6 ...~.)..N!.....
D593 1A FE 20 D8 32 0F D5 E5 CD 17 D6 3A 0F D5 D6 20 .. .2.....
D5A3 2A D2 D5 11 50 00 ED 4B D0 D5 04 0C E5 C5 77 23 *...P..K.....W
D5B3 10 FC 2B 77 19 0D 20 FB C1 E1 77 19 0D 20 FB B7 ..+W...W... ..
D5C3 ED 52 77 23 10 FC CD 17 D6 FB E1 F1 C9 00 00 00 .RwW.....
D5D3 00 00 CD A6 1A FE 50 30 F1 32 D1 D5 7E FE 2C 20 .....P0.2...~.,
D5E3 E9 23 CD A6 1A FE 18 30 E1 32 D0 D5 C9 01 50 08 .W.....0.2....P.
D5F3 3A D0 D5 5F 16 00 21 00 00 CB 39 30 01 19 CB 23 :...!....90...W
D603 CB 12 05 20 F4 11 00 F0 19 3A D1 D5 5F 16 00 19 ... ..!.....
D613 22 D2 D5 C9 F3 3A D4 D5 EE FF 32 D4 D5 D3 58 C9 *.....2...X.
    
```

This machine code routine is the 80 column version of the basic VPEEK and VPOKE. I have also added a box routine which might be usefull for titles and 'pretty screens'

The syntax for POKE is CMDPOKExx,yy,zz where xx is from 0 to 79, yy is from 0 to 23, and zz is the ASCII char: from 32 to 255.

Anything outside these limits will give an error, either ILLEGAL FUNCTION CALL or SYNTAX ERROR. The syntax for PEEK is...

CMDPEEKxx,yy: zz=PEEK(54545) where the above limits also apply, note that the byte to PEEK will only be correct if you assemble the routine at the location specified on the ORG statement.

Note also there are no brackets in the CMDPEEK.

The syntax for the box routine is CMDLINE(xx,yy)-(ii,jj) with an optional ,zz after it. If the zz is NOT specified, the last used value in either CMDPEEK or CMDLINE will be used. This means a box can be

drawn with a character taken from a specific screen location. In this case the zz=PEEK(54545) can be left out of the CMDPEEK command. The values ii,jj are absolute screen locations, not relative, & they must be positive in relation to xx,yy. If the total of xx plus ii or yy plus jj is more than 79 or 23 respectively, a funny box will result.Ok In effect, the CMDLINE command means 'draw a BOX with top/left location at xx,yy and bottom right location at ii,jj and use either the last character obtained with CMDPEEK, the character used in the last box or the character specified in the optional [zz]. Regardless of whether you use an assembler (ZEN) or poke the bytes in from the hex dump, when you save it to disk as a binary file use... BSAVE"1:filnam.ext",&HD503,&HD701

This is because the space used, in a spare area of the FDOS, is less than 2 pages (255 bytes) long, and BSAVE/BLOAD work in pages. Any bytes not used will be loaded as NOPs and will overwrite part of the FDOS. By saving 2 pages you will still overwrite when loading it back, but they will be the bytes that are supposed to be there, if you see what I mean ! Incidentally, the last byte specified in the BSAVE, i.e the end of your m/c routine will also be a nop if the total is not a multiple of 255 bytes so allways add one byte to the end in that case. Note that only some of the graphic characters can be used, from 160 to 191 and these have been shifted up the ASCII table to 224 to 255 because all the inverse characters, upper & lower case, are in there as well

As an example of what the CMDLINE can do, (since you may be wondering why I bothered to do it for a text screen) try this short basic program after you have the CMD commands on disk or in memory...

```
10 BLOAD"1:CMDxxxM/C",R
20 CLS: LOCATE28,11,0: PRINT"80 Col POKE, PEEK, & BOX
30 LOCATE26,12: PRINT"By Bob Curwood, Christchurch
40 A=-1: B=222: C=-1
50 C=C XOR -2:B=B+1 :IF B> 229 THEN 40
60 A=A+C: IF (A)-1 AND (A(11) THEN GOSUB 70: GOTO 40 ELSE 50
70 CMDLINE(A,A)-(79-(A#2),23-(A#2)),B
80 RETURN
```

If your monitor suffers from 'vision on sound' like mine, try turning the sound up while the above program is running !!

There still seem to be some obscure bugs in the CMD routines, if you come across any, please let me know all the symptoms.

## 80 COLUMN PEEK / POKE

### ADDITIONAL NOTES

By Jim Oliver

A little note on the 80 column peek / poke will be helpful to those less experienced folk.

The syntax for the poke is CMDPOKE COLUMN,LINE,ASCII CODE for example : CMDPOKE0,0,32 puts a space into the top LH corner. CMDPOKE79,23,65 puts a capital "A" into the bottom RH corner of the 80 column screen.

Similarly do a peek function like this : CMDPEEK0,0,ZZ=PEEK(54545) would return 32 for ZZ if the first poke example above had been done.

Either of these commands can be entered into a basic program so as to achieve the desired result. The values above can be used in the form of variables as well so there's no limit to the complexity of the use of these new commands. See the program below for a very simple but effective (and fast) 80 column screen print.

Thanks Bob for your efforts and we look forward to the BOX addition mentioned in your listing.

p.s The machine code program which is included in Bob's listing is available from P.D if you can't be bothered POKEing the values in.

```

10 REM 80 column peek and print. J.A.Oliver April 88 (WBVUG)
20 ON KEY GOSUB 90:KEY(1) ON
30 CLS
40 LOCATE 25,5:PRINT"Is "+CHR$(34)+"1:PEPO80"+CHR$(34)+" loaded
and run ?":A$=INPUT$(1):IFA$="Y"OR A$="y" THEN 60
50 BLOAD"1:PEPO80",R ' must be on this disc
60 LOCATE 10,10:PRINT"Now CLS and type required message onto the
80 column screen.                When you wish to print this
just hit F/key 1 then space bar"
70 SC$=INPUT$(1):PRINTSC$;GOTO 70
80 END
90 FOR Y=0 TO 22
100 FOR X=0 TO 79
110 CMDPEEKX,Y:LPRINTCHR$(PEEK(54545!));
120 NEXT X
130 NEXT Y
140 RETURN 80

```

### BRAINTEASER

We thought it would be an idea to start a 'brainteaser column' where we could present in each issue a different puzzle for you to program your way of. Any suggestions for actual puzzles will be gratefully accepted as will your "best solutions". We will judge these solutions and publish any really cunning ones. In the event of us not receiving any of your solutions (highly unlikely) we will publish our best solution. The programming language may be of any type with Basic preferred for the widest appeal.

Try this one : Recently I needed to make a oil drip tray to go under my car. What I needed was what a square tray with sides to give the maximum volume in case of an oil spill when changing my

oil. The sheet of metal I wanted to use was 40 centimeters square. All I wanted to do was cut 4 small corner pieces out and then fold up the sides and solder the joins, keeping the thing square. I can work down to an accuracy of half a mm.

What height was I to make the sides ?

.... Jim Oliver

.....

!!!!WANTED TO BUY!!!!

SV806 80 COLUMN CARD  
SV807 64K RAM CARTRIDGE  
SV902 DISK DRIVE  
SV903 DATA CASSETTE

...contact Bryan Fisk, 27 Karaka Street, Wainuiomata,  
Wellington.

.....

SUBSCRIPTIONS

\$18.00 only!

Encourage new members and ex-members to join

.....

BOOK REVIEW

Recently I came upon an excellent book for those of you who are interested in writing your own sort routines. It is :

Sorting Routines for Microcomputers  
By K. McLuckie & A. Barber  
Pub. Macmillan Press 1986  
(part of the Macmillan Microcomputer series)

This neat little book of around 150 pages is an excellent choice for the expert and beginner alike.

It begins by explaining that the data to be sorted is a very important starting point before deciding which sort to use. For example if the data is "ordered" in other words "nearly sorted" then the fastest sort by far is the simplest exchange sort. It takes only 4 seconds to sort 500 items of ordered data but takes 18 minutes for random data ! By comparison a real fast Quicksort

takes 48 seconds for 500 random and around the same time for ordered data. This demonstrates very well indeed that fast sort routines are not always fast; it depends on the data.

The main types of data are : random , ordered , reverse ordered , duplicated & interleaved.

The book starts out with exchange sorts and works through Shell-sorts (named after Mr. Shell), Quicksorts , binary tree sorts , heap sorts , tournament sorts , insertion sorts & others.

In all cases the sorts are compared as to times for the various data types. In each section of the book each sort is explained inside out and then each sort is 'optimised' in several steps of optimisation, each step carefully compared to the last.

In this way the reader is taken slowly through each sort routine until finally the optimum is attained. The Quicksort is a well known but little understood routine but this book raises the Quicksort up through several stages until finally there emerges a Hybrid QQSORT (Quick-Quicker-Sort) routine which is judged to be the best sort for every data type except the ordered variety where our old simple friend, the exchange sort is recommended.

QQSORT will sort 1500 items in about 150 seconds and the next fastest is the Shellsort at about twice that.

All the routines are developed in 'common Basic' ; BBC and Spectrum Basics (and one Pascal routine). Therefore they can be typed straight into SV. The ability to have SWAP as a command greatly enhances the sort speeds.

This book taught me why my Disk Directory Utility in newsletter Vol 4 No.2 was SO SLOW with ordered data and yet so fast with random data. I will now rewrite it so as to take advantage of the relative speeds of different sorts.

....Jim Oliver

.....

### FEEDBACK REQUIRED

In the mailbox we have had three requests for assistance. The first is from Jennifer Browne at Box 55, Tairua. She has an SV 3100 printer which she is having a few problems with. She is having trouble understanding all the control codes and would like to hear from anyone who has this printer up and running.

The second person is Bob Curwood (who contributed the 80 column POKE in this issue). Bob has an NEC 8001 computer and is interested in hearing from anyone who has one of these machines especially from the point of view of communications. The NEC has two interrupt driven RS 232 ports. Bob also has an ICL-15 PC which is an MP/M (a sort of multiuser CP/M) computer which supports discs with 36 system tracks !! The discs also have 1600 tracks with 4 sectors per track and two quad drives. And you thought SV was complex! Bob is thinking about starting up an RCPM using his ICL-15 since he says there are not enough BBS down here! Bob's address is : 124 Sullivan Ave, Christchurch 8002.

Lastly Jim Oliver wants to know how to put different fonts onto

SCREEN 1 so they can be screen dumped onto the printer. He has tried Elliot Young's PD program called "GRPlpt" without success.

If you can help any of these people please write direct to them or to the club's PD Box.

.....

### IS ANYONE DOING ANYTHING INTERESTING WITH

- GRAPHICS TABLETS
- CONVERSION OF PROGRAMS TO SVI
- PROGRAMING
- ROBOTICS
- STUNNING NEW ADVENTURE GAMES WITH SOUND AND GRAPHICS
- VOICE SYNTHESIS
- MODEMS

..... Write in for the next issue, become a published author!

.....

### Public Domain Update

The PD disc and tapes have now been expanded a little and the following are the details.  
This info formalises the listings placed in the Sept/Oct. Newsletter.

#### DISC No. 19

|           |  |                                        |
|-----------|--|----------------------------------------|
| FIVE      |  |                                        |
| PACK2     |  |                                        |
| INV       |  |                                        |
| MAIN      |  | ==> KING SOLOMONS MINES GAME           |
| MINES     |  | (excellent adventure game with         |
| PACK1     |  | graphics and sound)                    |
| THREE     |  |                                        |
| triffid   |  | catch the butterfly dodge triffid game |
| airattack |  | action game                            |
| bach1     |  | bach music                             |
| tenpin    |  | bowling / tenpin game                  |
| disk.dev  |  | Bryan Fisk's disk utility              |
| fildir.11 |  | disk directory & sort utility          |

#### DISC No. 20

This disc has all the recent Canadian PD files put onto it.

#### DISC No. 21

Don Stanley's programs now on to PD have been put onto Disc 21.

#### DISC / TAPE No. 22

|          |                            |
|----------|----------------------------|
| cannon   | howitzer shoot up game     |
| chess    | simple chess game          |
| hang     | hangman game               |
| invasion | invasion of Christchurch   |
| names    | club list utility (excell) |
| poker    | poker game                 |
| pont     | pontoon game               |
| sums     | simple arithmetic          |
| terror   | shoot and dodge game       |

#### DISC No. 23

|           |                           |
|-----------|---------------------------|
| catch8.8  | 80 column catch it game   |
| se23      | text screen to basic prog |
| PROOF1    | COMPUTE! proof typer      |
| revers4   | Othello game (fast)       |
| grab-em   | PAC MAN game              |
| cup       | football teams game       |
| solitaire | card game *               |
| monte2    | monte carlo cards *       |
| masterm   | detective who done it *   |

|         |                     |          |                       |
|---------|---------------------|----------|-----------------------|
| yahtzee | yahtzee game        | faceache | face drawing prog.    |
| bomb    | bomb the city game  | hikdoc   | tell the time prog.*  |
| latrom  | ROM key words list  | wari2    | Awari game *          |
| revers  | game of skill       | prtsr    | M/C text print screen |
| stark   | space invaders      | life40   | Game of life          |
| randf   | random file demo    | vpoke    | vpoked fonts (clever) |
| alpha   | alphabet game       | DAIRY    | diary keeper          |
| screen  | screen save utility | scroll   | M/C text scroll       |
|         |                     | mod.bas  | graphics demo *       |
|         |                     | kaleid   | kaleidoscope colors * |
|         |                     | tennis   | simple tennis game    |
|         |                     | qplott   |                       |
|         |                     | conesv   | ==> 3D plotters       |
|         |                     | parabo   | (graphics)            |
|         |                     | hyperb   |                       |
|         |                     | plotter  | 3D plot of functions  |

Disc 22 contains programs from Leister Reilly in Christchurch and are all of excellent quality. The programs on Disc 23 with an "\*" beside them are from COMPUTE! magazine and are generally of a professionally high quality.

### DISC No.24

This disc was generously sent to us by Bob Curwood from Christchurch. It contains some of the best program protection schemes ever devised for SV. The challenge is up to you, break down all the protections used AND get an excellent circuit drawing utility as well. This will enable you to draw almost any electronic symbol and to place them into a circuit drawing on screen. This can be dumped to a printer or saved to disc. There are 40 odd sprites used (resistors, capacitors, IC's, transformers etc,etc) and you can add your own. This disc contains some high quality stuff which will keep you amused and amazed for hours.

You will note that some of the PD programs are on disc only. In these cases the programs are best or will run from disc only. This does not mean that this stuff will not run from tape AT ALL, but it does mean that you will need some expertise to sort it out. Some of the best games (King Solomon's Mines) will (in theory) run from tape and I may consider converting it to tape use if there is enough interest.

Please remember that maintaining a PD library is of interest to me but it is sometimes a long and time consuming job which requires a good deal of dedication and patience. If you have anything which you consider would be of interest to others then please share it with us. There are now 24 discs and about 22 tapes of PD stuff in our library so if you are trying to write a program then it's fairly likely we have something for you!

Costs for this service are trivial at \$3.00 for copying each disc or tape plus \$2.50 for postage and handling on each order. Remember that we will supply the tapes but the discs need to come from you and formatted on your drives so as to prevent any track alignment problems.

Send your requests to SV P.D SOFTWARE  
PO BOX 26050  
NEWLANDS, WELLINGTON

NEW ZEALAND

.... Jim Oliver

.....  
ANYONE HAVE A

- QUESTION

- PROBLEM

- COMMENT

- REQUEST

..... take matters into your own hands and write in now!!!!

To the Editor,

I enjoy reading the club magazine and so this is my contribution. The program makes the cursor shape change constantly. This is great for livening up the input sections on programs where you have to input a lot of information.

I have a BVI-328 with the old metal expander and

two disk drives, both single sided. I use an amber monitor. Inside the expander I have the centronics interface and the 80 column card.

I am willing to provide "The Magic of Spectravideo" on disk to anyone who provides proof of purchase of the tape -- and \$8 for the (2) disks and postage.

Yours in computing  
Michael Warwick.

CURSOR PROGRAM

This program makes the cursor into a constantly changing shape. It can be easily changed to allow for an almost infinite number of moving shapes. Enter in the following program to see what I mean.

```
10 N=13:I=1
20 CLS:DIMA(N):GOSUB1000
30 ON INTERVAL=10 GOSUB2000
40 INTERVAL ON
50 PRINT "LIKE THE CURSOR ? :";
60 A$=INKEY$
70 IF A$="" THEN GOTO 60
80 PRINT A$
90 IF A$="Y" THEN PRINT "Yes, It's GREAT !!"

100 IF A$="N" THEN PRINT "Wow, that's what I call amazing!! "
110 GOTO 110

1000 RESTORE 1040: FOR X= 1 TO N
```

```
1010 READ A
1020 A(X)=A
1030 NEXT X
1040 DATA 16,56,40,56,16,56,40,56,16,56
1050 DATA 40,56,16
1060 RETURN
```

```
2000 FOR X=3576 TO 3583
2010 I=I+1: IF I=N+1 THEN I=1
2020 VPOKE X,A(I)
2030 NEXT X
2040 I=I-5: IF I<1 THEN I=N+1: RETURN
```

Run it (if you haven't already), and watch the cursor. Amazing ? not quite. When the cursor moves, like when something is printed onto the screen, the cursor changes back to its old self. Also the computer checks for INTERVAL between commands, so the INPUT command stops the movement of the pattern, and the cursor changes back after the first letter is pressed.

Now stop the program and try another pattern by adding the following:

```
10 N=38: I=0

1040 DATA 16,56,40,56,16,56,40,56,16,56,40,56,
          16,56,40,56,16,16,16,16

1050 DATA 16,84,124,84,56,0,84,168,0,168,84,0,
          84,168,0,168,84,0
```

Pretty neat huh? But unless it is a program featuring the Navy, this isn't a very good pattern for use when writing programs. What we need is an eye catching pattern which fits into almost any program.

Stop the program and try this pattern:

```
10 N=20: I=1

1040 DATA 48,96,192,192,96,48,24,12,12,24

1050 DATA 48,96,192,192,96,48,24,12,12,24
```

Not to your taste? Try the next one:

```
10 N=9: I=1

1040 DATA 36,72,144,36,72,144,36,72,144

1050 REM FOR MORE DATA
```

When you make your own patterns remember that you need at least seven data elements, or else change line 2040.

.....

WELCOME TO THE FOLLOWING NEW, RENEWED, TRANSFERRED MEMBERS  
NICE TO HAVE YOU ON BOARD

|                     |                |
|---------------------|----------------|
| Doug Pratt          | - Wellington   |
| David Roberts       | - Australia    |
| Henry Ohia          | - Wellington ? |
| R. Moore            | - Australia    |
| E.N. Cowan          | - Wanganui     |
| R.E & T.A. Phillips | - Wellington   |

.....

```

10  *****
20  *****
30  ***                               **
40  *** 80COL PEEK/POKE ***
50  ***      by ***
60  ***  Bob Curwood ***
70  ***                               **
80  *****
90  *****
100
110  !The syntax for poke is CMDPOKExx,yy,zz
120
130  !The syntax for peek is CMDPEEKxx,yy;zz=PEEK(34545)
140
150  !In both cases, the parameters can be in any numerical form,
160  ie.g &H1234,&0123,&B10010, or variables I,J,K, A+B/C
170
180  !The routine is placed in an area of the FDOS not used after
190  !boot-up to keep it clear of a basic program. You can of
200  !course assemble it anywhere you like but note the byte to
210  !PEEK after a CMDPEEK will need to be changed.
220
230  !The lowest ASCII character that can be POKEed/PEEKed is a
240  !space, (32 dec). Both normal & inverse alphanumeric can be
250  !used but only a restricted graphic set. Dont blame me for
260  !this!
270
280  !I am also working on a small addition to this routine that
290  !will place a character box on the crt (just like the title
300  !above) using similar syntax to the the SVI box, and either
310  !the last obtained zz from CMDPEEK or as specified in the
320  !command. Syntax for this will be CMDLINE(xx,yy)-(ii,jj)[,zz]
330  !with xx,yy as above and ii,jj a positive displacement.
340  !The zz of course is the character to be used.
350  !The first version of this is FAST!, much quicker than the
360  !SVI box.
370
380  !The routine must be initialised either by BLOADING with an ,R
390  !after the filename, or DEFUSR=&HD503:A=USR(0)

```

```

400
410      ORG 0D503H
420      LOAD 0D503H
430      GETBYT: EQU 1AA6H
440      OFFSET: EQU 0F000H
450      SNERR:  EQU 08EDH
460      POKTOK: EQU 152
470      PEKTOK: EQU 151
480      ALTHOK:  LD HL,OFF9CH
490                LD A,0C3H
500                LD DE,START
510                LD (HL),A
520                INC HL
530                LD (HL),E
540                INC HL
550                LD (HL),D
560                RET
570      SAVPEK:  DB 0
580      START:   LD A,(HL)
590                CP POKTOK
600                JR Z,POKE

```

```

610                CP 255
620                RET NZ
630                RST 10H
640                CP PEKTOK
650                RET NZ
660      PEEK:     INC HL
670                CALL LOCATE
680                PUSH HL
690                LD HL,(SAVLOC)
700                CALL CHGBNK
710                LD E,(HL)
720                CALL CHGBNK
730                LD A,32
740                ADD A,E
750                LD (SAVPEK),A
760                EI
770      EREXIT:   POP HL
780      EXIT:     POP AF
790                RET
800      POKE:     INC HL
810                CALL LOCATE
820                LD A,(HL)
830                PUSH HL
840                RST 8H
850                DB ','
860                POP HL
870                INC HL
880                CALL GETBYT
890                CP ' '
900                RET C
910                SUB 32
920                PUSH HL
930                LD E,A
940                CALL CHGBNK
950                LD A,E
960                LD HL,(SAVLOC)

```

|      |         |              |      |         |               |
|------|---------|--------------|------|---------|---------------|
| 970  |         | LD (HL),A    | 1210 |         | LD D,0        |
| 980  |         | CALL CHGBNK  | 1220 |         | LD HL,0000    |
| 990  |         | POP HL       | 1230 | SAVLP:  | BRL C         |
| 1000 |         | EI           | 1240 |         | JR NC,BYPASS  |
| 1010 |         | JR EXIT      | 1250 |         | ADD HL,DE     |
| 1020 | SAVEX:  | DW 0000      | 1260 | BYPASS: | SLA E         |
| 1030 | SAVLOC: | DW 0000      | 1270 |         | RL D          |
| 1040 | LOCATE: | CALL GETBYT  | 1280 |         | DEC B         |
| 1050 |         | CP 0         | 1290 |         | JR NZ,SAVLP   |
| 1060 |         | JR C,EXIT    | 1300 |         | LD DE,OFFSET  |
| 1070 |         | CP 80        | 1310 |         | ADD HL,DE     |
| 1080 |         | JR NC,EXIT   | 1320 |         | LD DE,(SAVEX) |
| 1090 |         | LD (SAVEX),A | 1330 |         | ADD HL,DE     |
| 1100 |         | LD A,(HL)    | 1340 |         | LD (SAVLOC),H |
| 1110 |         | RST 8H       | 1350 |         | POP HL        |
| 1120 |         | DB ','       | 1360 |         | RET           |
| 1130 |         | CALL GETBYT  | 1370 | TOGGLE: | DB 0          |
| 1140 |         | PUSH HL      | 1380 | CHGBNK: | DI            |
| 1150 |         | CP 0         | 1390 |         | LD A,(TOGGLE) |
| 1160 |         | JR C,EREXIT  | 1400 |         | XOR 255       |
| 1170 |         | CP 24        | 1410 |         | LD (TOGGLE),A |
| 1180 |         | JR NC,EREXIT | 1420 |         | OUT (88),A    |
| 1190 | MULTI:  | LD BC,0850H  | 1430 |         | RET           |
| 1200 |         | LD E,A       | 1440 | END     |               |

|      |                                                 |                 |
|------|-------------------------------------------------|-----------------|
| D503 | 21 9C FF 3E C3 11 12 D5 77 23 73 23 72 C9 80 7E | !..>...wWwW..~  |
| D513 | FE 98 28 20 FE FF C0 D7 FE 97 C0 23 CD 5E D5 E5 | ..( .....W.^..  |
| D523 | 2A 5C D5 CD 9F D5 5E CD 9F D5 3E 20 83 32 11 D5 | *\.....^> .2..  |
| D533 | FB E1 F1 C9 23 CD 5E D5 7E E5 CF 2C E1 23 CD A6 | ....W.^~...W..  |
| D543 | 1A FE 20 D8 D6 20 E5 5F CD 9F D5 7B 2A 5C D5 77 | .. .. _...(\.w  |
| D553 | CD 9F D5 E1 FB 18 DB 4F 00 7F F7 CD A6 1A FE 00 | .....O.....     |
| D563 | 38 D0 FE 50 30 CC 32 5A D5 7E CF 2C CD A6 1A E5 | 8..P0.2Z.~..... |
| D573 | FE 00 38 BD FE 18 30 89 01 50 08 5F 16 00 21 00 | ..8...0..P...1. |
| D583 | 00 CB 39 30 01 19 CB 23 CB 12 05 20 F4 11 00 F0 | ..90...W... ..  |
| D593 | 19 ED 5B 5A D5 19 22 5C D5 E1 C9 00 F3 3A 9E D5 | ..{Z...".....   |
| D5A3 | EE FF 32 9E D5 D3 58 C9                         | ..2...X.        |

PAGE 1

|        |             |             |             |      |
|--------|-------------|-------------|-------------|------|
| ALTHOK | D503 BYPASS | D589 CHGBNK | D59F EREXIT | D534 |
| EXIT   | D535 GETBYT | 1AA6 LOCATE | D55E MULTI  | D57B |
| OFFSET | F000 POKTOK | 0098 PEKTOK | 0097 PEEK   | D51E |
| POKE   | D537 SNERR  | 08ED SAVPEK | D511 START  | D512 |
| SAVEX  | D55A SAVLOC | D55C SAVLP  | D584 TOGGLE | D59E |

NOW SEE

Jim's NOTES on page 7 *JP*